Piper PA-46R-350T Matrix (MATRIX)

Lycoming TIO-540-AE2A, turbocharged and intercooled, 350 hp · 6 seats · Typical PA-46R-350T Matrix (2008+, TIO-540-AE2A, 4,340 lb, unpressurised Mirage airframe, hydraulic gear) POH procedures. Written for either the Avidyne Entegra (2008–10, S-TEC 55X) or Garmin G1000 (2010+, GFC 700) panel; ice protection and built-in oxygen where fitted.

V-speeds & limits

Vr	**70–75 KIAS (liftoff)**
Vx	**85 KIAS**
Vy	**110 KIAS**
Best glide	**approx 95 KIAS (gear & flaps up, prop low RPM)**
Vfe	**165 KIAS (10°) / 130 KIAS (20°) / 116 KIAS (36° full)**
Vlo	**165 KIAS extend / 130 KIAS retract**
Vle	**165 KIAS**
Va	**133 KIAS (4,340 lb)**
Vno	**170 KIAS**
Vne	**198 KIAS**
Vs / Vso	**approx 69 / 59 KIAS (max weight)**
Vref	**80–85 KIAS final (full flaps); approx 78 KIAS short field**
Emergency descent	**165 KIAS gear down / 133 KIAS in turbulence**

Normal procedures

Before Start

- Preflight inspection ..... **COMPLETE (fuel sumps, deice boots and static wicks, tyres, cowl and turbo area, oxygen bottle pressure)**
- Weight & balance / takeoff data ..... **COMPUTED (4,340 lb max takeoff)**
- Passenger briefing ..... **COMPLETE (belts, airstair door, emergency exit, oxygen masks / cannulas, no smoking)**
- Seats, belts, shoulder harnesses ..... **ADJUST & LOCK**
- Cabin door (upper & lower) / baggage doors ..... **CLOSED & LATCHED (DOOR AJAR annunciator out once power is on)**
- Parking brake ..... **SET**
- Landing gear selector ..... **DOWN**
- Emergency gear extension lever ..... **NORMAL / STOWED (floor, between front seats)**
- Throttle / propeller / mixture ..... **CLOSED / FULL FORWARD / IDLE CUTOFF**
- Fuel selector ..... **FULLEST TANK (LEFT or RIGHT – OFF is detented, there is no BOTH)**
- Avionics master ..... **OFF**
- Battery master ..... **ON – voltage CHECK (approx 24 V)**
- Fuel quantity ..... **CHECK both tanks (120 gal usable total)**
- Annunciators / PFD alerts ..... **TEST / REVIEW (gear lights three green)**
- Standby instruments / backup battery ..... **TEST (standby attitude indicator erect, backup battery test passes)**
- Oxygen ..... **CHECK pressure (approx 1,850 psi full) and duration for planned altitude; masks / cannulas available**
- Alternate static source ..... **OFF (normal)**
- Beacon ..... **ON**

Engine Start

- Propeller area ..... **CLEAR**
- Throttle ..... **APPROX 1/2 in OPEN**
- Propeller ..... **FULL FORWARD (high RPM)**
- Mixture ..... **FULL RICH**
- Fuel pump ..... **HI until fuel flow indicates (approx 3–5 sec prime), then OFF**

▶ **Engine hot or flooded?**

No:

Yes:

- Mixture ..... **IDLE CUTOFF**
- Throttle ..... **FULL OPEN**
- Mixture ..... **ADVANCE to FULL RICH as engine fires, throttle to idle**

- Starter ..... **ENGAGE (release when engine fires; 30 sec max, then cool)**
- Throttle ..... **1,000 RPM**
- Oil pressure ..... **25 psi within 30 sec (normal 55–95 psi)**
- Alternator 1 & 2 switches ..... **ON – ALT 1 / ALT 2 INOP annunciators out, voltage approx 28 V**
- Fuel pump ..... **OFF (LO if fuel flow fluctuates)**
- Avionics master ..... **ON**
- PFD / MFD ..... **CHECK (AHRS aligned, no red X's, engine page displayed, database current)**
- Engine instruments ..... **CHECK (oil pressure, oil temp rising, fuel flow, TIT, CHT)**

Before Taxi

Brakes	**CHECK on initial roll**
PFD / MFD / standby instruments	**CHECK & SET (altimeters agree, heading, attitude erect, HSI turns correctly in turns)**
Flight plan / avionics	**LOADED & SET**
Autopilot / electric trim	**PREFLIGHT TEST – disconnect checks, trim both directions – OFF**
Pitot heat / stall warning heat	**ON momentarily – CHECK (annunciator out, load rise), then AS REQUIRED**
Propeller deice	**ON – PROP HEAT indication, then AS REQUIRED (if fitted)**
Windshield heat	**ON – CHECK, then AS REQUIRED (if fitted)**
Surface deice (boots)	**SINGLE CYCLE – boots inflate / deflate, pressure indication (if fitted)**
Elevator / rudder trim	**SET for takeoff**
Fuel selector	**CHECK fullest tank (LEFT or RIGHT)**
Flaps	**SET for takeoff (0° normal; 10° short field) – indicator CHECK**

Before Takeoff

Parking brake	**SET**
Flight controls	**FREE & CORRECT**
Mixture	**FULL RICH**
Throttle	**2,000 RPM**
Magnetos	**CHECK L & R (175 RPM max drop, 50 RPM max difference)**
Propeller	**CYCLE 2–3 times (RPM drops, then returns)**
Engine instruments	**CHECK (oil pressure 55–95 psi, oil temp in green, fuel flow steady, TIT / CHT)**
Alternators	**CHECK both charging, loads shared, voltage approx 28 V**
Throttle	**IDLE check, then 1,000 RPM**
Fuel selector	**FULLEST TANK (feel for detent)**
Fuel pump	**LO**
Oxygen	**SET for climb (ON above 12,500 ft; masks / cannulas connected if cruising high)**
Propeller	**FULL FORWARD**
Flaps	**SET & INDICATING (0° or 10°)**
Trim	**SET takeoff range**
Landing gear selector	**DOWN – three green**
Ice protection / pitot heat	**AS REQUIRED**
Lights	**AS REQUIRED**
Transponder	**ALT**
Doors & windows	**LATCHED (DOOR AJAR out)**
PFD / MFD	**NO ALERTS; standby instruments CHECK**
Takeoff briefing	**COMPLETE (abort point; throttle slowly to 42 in Hg / 2,500 RPM, fuel flow approx 36–38 GPH; gear up with positive rate and no usable runway, below 130 KIAS)**

After Takeoff / Climb

Takeoff power	**FULL THROTTLE applied slowly – 42 in Hg max, 2,500 RPM, fuel flow approx 36–38 GPH**
Landing gear	**UP (positive rate, no usable runway, below 130 KIAS) – lights out, HYD PUMP light out**
Flaps	**UP (above approx 85 KIAS, obstacles cleared)**
Climb power	**REDUCE SLOWLY to approx 35 in Hg / 2,500 RPM (cruise climb) above 1,000 ft AGL**
Mixture	**FULL RICH / climb fuel flow per POH schedule (TIT below 1,650°F)**
Fuel pump	**OFF above 1,000 ft AGL (LO if fuel flow fluctuates)**
Airspeed	**110 KIAS (Vy) / 120–130 KIAS cruise climb for cooling**
Oxygen	**ON passing 12,500 ft – flow checked at each station (masks above 18,000 ft)**
Engine instruments	**CHECK (CHT, TIT, oil temp 245°F max, oil pressure)**
Ice protection	**AS REQUIRED**
Landing / taxi lights	**OFF**

Cruise

Cruise power	**SET (approx 2,400 RPM / 30–32 in Hg, 75% max recommended) – change power slowly**
Mixture	**LEAN SLOWLY per POH cruise table – TIT 1,650°F max recommended cruise (1,750°F red line)**
Fuel selector	**ALTERNATE LEFT / RIGHT to keep tanks balanced – note time**
Fuel pump	**LO if fuel flow fluctuates (vapour suppression at altitude / hot fuel), otherwise OFF**
Fuel quantity / flow	**CHECK against plan (MFD fuel totaliser cross-checked with gauges)**
Oxygen	**CHECK flow at each station, pressure and remaining duration; pulse oximeter as available**
Engine instruments	**CHECK (oil 55–95 psi / 245°F max, CHT, TIT)**
Ice protection	**AS REQUIRED**
PFD / MFD	**CHECK (no alerts, altimeter set, heading / GPS agree)**

Descent

Altimeters	**SET (PFD and standby)**
Power	**REDUCE SLOWLY (approx 1 in Hg per minute) – avoid rapid cooling**
Mixture	**ENRICH progressively (TIT / CHT monitored)**
Fuel selector	**FULLEST TANK (LEFT or RIGHT)**
Oxygen	**OFF below 12,500 ft (after cannulas / masks removed)**
Windshield heat / ice protection	**AS REQUIRED**
Airspeed	**BELOW 170 KIAS (Vno); plan below 165 KIAS before gear extension**
Approach briefing	**COMPLETE (Vref 80–85, flap setting, go-around power applied slowly)**
Seat belts / harnesses	**SECURE**

Before Landing

Fuel selector	**FULLEST TANK (LEFT or RIGHT)**
Fuel pump	**LO**
Mixture	**FULL RICH (as required at high field elevation)**
Airspeed	**BELOW 165 KIAS**
Landing gear	**DOWN – three green, HYD PUMP light out**
Flaps	**10° below 165 KIAS; 20° below 130 KIAS; 36° on final below 116 KIAS**
Propeller	**FULL FORWARD (on final)**
Autopilot / yaw damper	**OFF**
Airspeed	**80–85 KIAS final (full flaps; approx 78 KIAS short field; add 10–15 KIAS with residual ice)**
Landing light	**ON**
Landing gear	**THREE GREEN – CONFIRMED on final**

After Landing

Flaps	**UP**
Fuel pump	**OFF**
Ice protection	**OFF (pitot heat, prop deice, windshield heat, boots)**
Landing light / strobes	**AS REQUIRED**
Transponder	**STBY / as required (GND on G1000)**
Trim	**SET takeoff**

Shutdown

Parking brake	**SET**
Throttle	**1,000 RPM – turbocharger cool-down 3–5 min (TIT stabilised / falling)**
Avionics master	**OFF**
Electrical equipment / lights	**OFF**
Oxygen	**OFF**
Mixture	**IDLE CUTOFF**
Magnetos	**OFF (after propeller stops)**
Alternator 1 & 2 / battery master	**OFF**
Control lock	**INSTALL**
Chocks / tiedowns / covers	**AS REQUIRED**

Abnormal procedures

Alternator 1 / 2 Failure

ALT 1 INOP or ALT 2 INOP annunciator, load shift to one alternator, or low voltage

Affected alternator switch	**OFF, then ON (reset)**
Alternator field circuit breaker	**CHECK / RESET once**

▶ Alternator restored?

Yes:

Alternator loads / voltage	**MONITOR both**

No – one alternator failed:

Affected alternator switch	**OFF**
Electrical load	**REDUCE within the remaining alternator's capacity (windshield heat, prop deice, air conditioning OFF as needed)**
Deferred: One alternator inoperative – electrical load limited, no flight into icing; land as soon as practical.	**NOTED**

No – both alternators failed:

Both alternator switches	**OFF**
Non-essential electrical load	**OFF (battery only – approx 30 min; one display, one nav-com, transponder, pitot heat)**
Standby instruments / backup battery	**CONFIRM available**
Landing gear	**EXTEND EARLY while battery power remains (free-fall extension if the pump fails)**
Deferred: Both alternators failed – battery only; extend gear early, flaps may fail, displays may go dark – standby instruments; land as soon as possible.	**NO**

Low Oil Pressure

Oil pressure below 55 psi at cruise power (25 psi min idle)

Oil pressure / oil temperature ..... **CHECK and cross-check (engine page / standby indication)**

▶ Oil temperature rising or pressure below 25 psi?

Yes – genuine:

- Power ..... **REDUCE to minimum required (turbocharger is oil-lubricated – expect failure)**
- Airport ..... **DIVERT to nearest suitable, stay within glide of a landing area**
- Altitude ..... **HOLD altitude as long as practicable for glide range**
- Deferred: Low oil pressure – land as soon as possible, expect possible engine failure. ..... **NOTED**

No – gauge suspect:

- Flight ..... **CONTINUE, monitor oil temperature and pressure**
- Deferred: Oil pressure indication suspect – maintenance check required. ..... **NOTED**

Turbocharger Failure / Manifold Pressure Loss

Manifold pressure falls or cannot be obtained, fuel flow high for the MP, rough running, possible exhaust smell

Throttle ..... **REDUCE, then re-set smoothly**

Mixture ..... **LEAN to smooth running at the manifold pressure available (fuel flow too high for MP)**

Engine instruments ..... **CHECK (TIT, CHT, oil pressure and temperature)**

Exhaust leak indications ..... **CHECK (smell, smoke, cabin CO) – exhaust gas in cowling is a fire hazard**

▶ Exhaust leak or fire suspected?

Yes:

- Power ..... **MINIMUM required**
- Cabin heat ..... **OFF (CO risk)**
- Landing ..... **NEAREST SUITABLE AIRPORT – be ready for Engine Fire in Flight**
- Deferred: Turbocharger / exhaust failure – fire hazard, land as soon as possible, no go-around. ..... **NOTED**

No – power loss only:

- Altitude ..... **DESCEND to where naturally-aspirated power is adequate (approx 28 in Hg available near sea level, less with altitude)**
- Oxygen ..... **AS REQUIRED while descending from high altitude**
- Deferred: Turbocharger failed – reduced power, low altitude only, longer takeoff distance; land as soon as practical. ..... **NOTED**

High TIT / Overboost

TIT above 1,750°F red line or manifold pressure above 42 in Hg

▶ Indication?

Overboost (MP above 42 in Hg):

- Throttle ..... **REDUCE below 42 in Hg, re-apply slowly**
- Oil temperature ..... **CHECK (cold oil slows the density controller – warm up before full power)**
- Momentary overboost of approx 1 in Hg for a few seconds is tolerated; sustained or large overboost requires a maintenance inspection before further flight.
- Deferred: Overboost occurred – record peak MP and duration, maintenance inspection required. ..... **NOTED**

High TIT:

- Mixture ..... **ENRICH until TIT below 1,650°F**
- Throttle ..... **REDUCE if TIT remains high**
- Fuel flow ..... **CHECK against POH for the power set (low flow – fuel pump LO)**

▶ TIT within limits?

Yes:

- Flight ..... **CONTINUE, richer mixture**

No:

- Power ..... **REDUCE to the highest setting that keeps TIT below 1,650°F**
- Deferred: High TIT – reduced power, suspect fuel system or TIT probe; maintenance check. ..... **NOTED**

Fuel Flow Fluctuation / Low Fuel Pressure

Fuel pump ..... **LO (vapour suppression); HI only if fuel flow is very low – may flood engine at low power**

Fuel selector ..... **SWITCH TANKS (check quantity in the tank being used)**

Mixture ..... **ADJUST for smooth running**

Engine instruments ..... **CHECK (fuel flow, TIT, CHT)**

▶ Fuel flow steady and engine smooth?

Yes:

- Deferred: Fuel flow fluctuation cleared – leave fuel pump LO; report to maintenance. ..... **NOTED**

No:

- Power ..... **REDUCE to smoothest setting; lower altitude helps vapour**
- Deferred: Fuel flow unstable – land as soon as practical at nearest suitable airport. ..... **NOTED**

Landing Gear Unsafe / Fails to Retract or Extend

Red gear warning light, fewer than three green, or HYD PUMP light stays on

Airspeed ..... **BELOW 165 KIAS (below 130 KIAS to retract)**

Gear indicator bulbs ..... **CHECK (press to test) – navigation lights dim the gear lights**

Hydraulic pump circuit breaker ..... **CHECK / RESET once**

Landing gear selector ..... **CYCLE once to the desired position**

▶ Gear in desired position?

Yes:

Gear indication ..... **CONFIRM three green / lights out**

Deferred: Gear system fault – do not retract after landing; hydraulic / gear inspection required. ..... **NOTED**

No – failed to retract:

Landing gear selector ..... **DOWN – leave down**

Hydraulic pump ..... **If HYD PUMP light stays on, PULL hydraulic pump circuit breaker (pump running against a leak)**

Airspeed ..... **BELOW 165 KIAS**

Deferred: Gear will not retract – gear down, 165 KIAS max, reduced range; land as soon as practical. ..... **NOTED**

No – failed to extend:

Use Emergency Gear Extension (Free Fall).

Flap Failure / Asymmetry

Flap selector ..... **STOP at current position – do not continue extension**

Flap circuit breaker ..... **CHECK / RESET once**

▶ Flaps operable and symmetric?

Yes:

Flap setting ..... **AS REQUIRED**

No:

Landing ..... **AT CURRENT FLAP SETTING – approach approx 90–95 KIAS flaps up, longer runway**

Deferred: Flaps inoperative / asymmetric – flaps-up or partial-flap landing, approach approx 90–95 KIAS, increased landing distance. ..... **NOTED**

Propeller Overspeed

RPM above 2,500 and not responding to the prop lever

Throttle ..... **REDUCE**

Propeller ..... **FULL AFT (low RPM)**

Airspeed ..... **REDUCE (pitch up) to bring RPM within limits**

Oil pressure ..... **CHECK (governor uses engine oil)**

Power ..... **SET minimum that keeps RPM at or below 2,500**

Deferred: Propeller governor fault – control RPM with throttle and airspeed, land as soon as practical. ..... **NOTED**

Door Open in Flight

DOOR AJAR annunciator or door noise / buffet

Aircraft ..... **CONTROLLABLE – fly the airplane**

Airspeed ..... **REDUCE to approx 110–120 KIAS**

Belts / loose items ..... **SECURE; passengers away from the door**

Do not attempt to close or re-latch the door in flight. Expect noise and buffet; the upper door may trail open.

Deferred: Cabin door unlatched – return and land as soon as practical, normal approach speed. ..... **NOTED**

Oxygen Quantity Low / No Flow

Oxygen pressure low, flow indicator not showing flow, or duration insufficient for planned cruise

Oxygen bottle valve / system switch ..... **CHECK ON**

Masks / cannulas / connections ..... **CHECK seated, lines not kinked, flow indicators**

Remaining duration ..... **COMPUTE for occupants at current altitude**

▶ Oxygen sufficient to continue above 12,500 ft?

Yes:

Flow ..... **MONITOR every 15 min; descend before pressure reaches minimum**

Deferred: Oxygen quantity limited – descend below 12,500 ft before supply exhausted. ..... **NOTED**

No:

Descent ..... **INITIATE to 12,500 ft or below (10,000 ft with passengers) before supply runs out**

Crew ..... **MONITOR for hypoxia symptoms (Hypoxia procedure)**

Deferred: Oxygen exhausted / failed – remain at or below 12,500 ft, replan route and altitude. ..... **NOTED**

MFD / AHRS / Air Data Failure

MFD blank, AHRS alignment lost (red X attitude / heading), or ADC failure (red X airspeed / altitude)

▶ Failed unit?

MFD:

- Display reversion — **SELECT (G1000: DISPLAY BACKUP – engine data on PFD; Avidyne: PFD and standby engine indications)**
- Engine monitoring — **USE reversionary engine strip; set power conservatively**
- Navigation — **CONTINUE on PFD / second GPS-nav-com**
- Deferred: MFD failed – reversionary engine display, reduced moving-map / traffic / weather; maintenance required. — **NOTED**

AHRS:

- Standby attitude indicator / compass — **FLY attitude and heading**
- Autopilot — **OFF (attitude source lost); re-engage only with valid attitude**
- AHRS re-alignment — **ALLOW – straight and level, wings level, approx 1 min**
- IMC — **EXIT / AVOID until attitude restored**
- Deferred: AHRS failed – standby attitude indicator, no autopilot, visual approach preferred. — **NOTED**

Air data (ADC):

- Standby airspeed / altimeter — **FLY from standby instruments**
- Pitot heat — **ON (icing may be the cause)**
- Autopilot altitude / speed modes — **DO NOT USE; transponder altitude may be lost – ATC advised**
- Deferred: Air data failed – standby airspeed / altimeter, no altitude reporting; approach by attitude and power. — **NOTED**

Pitot / Static Source Blockage

Airspeed erratic or zero, altimeter / VSI frozen (PFD and standby agree, so not a display fault)

- Pitot heat — **ON**
- Attitude and power — **FLY KNOWN pitch / power settings; GPS ground speed as cross-check**

▶ Static system blocked?

Yes:

- Alternate static source — **ON (cabin); expect small altimeter / airspeed errors; vents and storm window closed**
- Deferred: Static source blocked – alternate static in use, altimeter error on approach. — **NOTED**

No – pitot only:

- Airspeed indicator — **DISREGARD until heat clears the blockage; use attitude, power and GPS**
- Deferred: Airspeed unreliable – approach by attitude and power, add margin, longer runway. — **NOTED**

Icing Conditions / Ice Accumulation

Ice forming on wing leading edge / ice light inspection

- Pitot heat / stall warning heat — **ON**
- Propeller deice — **ON (if fitted)**
- Windshield heat — **ON (if fitted)**
- Surface deice (boots) — **CYCLE at first sign of ice – repeat as required (if fitted)**
- Autopilot — **DISCONNECT periodically – check for trim / control changes**
- Exit icing — **CLIMB / DESCEND / TURN as required (aircraft without ice protection must leave icing immediately)**

▶ Residual ice on airframe for landing?

No:

- Ice protection — **OFF when clear**

Yes:

- Flaps — **10° maximum for landing**
- Approach speed — **INCREASE 10–15 KIAS; no abrupt pitch changes**
- Deferred: Residual airframe ice – flaps 10°, approach 95–100 KIAS, longer landing distance. — **NOTED**

Emergency procedures

Engine Failure During Takeoff Roll

- Throttle (memory) — **CLOSED**
- Brakes (memory) — **AS REQUIRED – stop straight ahead**
- Mixture — **IDLE CUTOFF**
- Fuel selector — **OFF**
- Magnetos — **OFF**
- Battery master / alternators — **OFF**

Engine Failure After Takeoff

Airborne at low altitude

LAND IMMEDIATELY

- Airspeed (memory) ..... **95 KIAS (best glide, gear & flaps up); 80–85 KIAS with gear / flaps down**
- Fuel pump (memory) ..... **HI (if fuel flow low); fuel selector SWITCH TANKS if time permits**

▶ Runway remaining to land on?

Yes:

- Landing gear ..... **DOWN (if retracted)**
- Flaps ..... **AS REQUIRED**
- Land ..... **STRAIGHT AHEAD**

No – land ahead off-field:

- Landing gear ..... **DOWN on firm surface / UP on rough terrain or water**
- Flaps ..... **DOWN (36°) when landing assured**
- Fuel selector ..... **OFF**
- Mixture ..... **IDLE CUTOFF**
- Magnetos ..... **OFF**
- Battery master ..... **OFF (after gear position set)**

Land straight ahead with only small changes of direction. Do not attempt a turn back.

Engine Failure in Flight

Loss of power with altitude available

LAND AS SOON AS POSSIBLE

- Airspeed (memory) ..... **95 KIAS (best glide, gear & flaps up)**
- Fuel selector (memory) ..... **SWITCH TANKS (feel for detent)**
- Fuel pump (memory) ..... **LO; HI if fuel flow reads zero (lean if engine floods)**
- Mixture (memory) ..... **FULL RICH, then lean as required**
- Propeller (memory) ..... **FULL AFT (low RPM) to extend glide**

Oxygen ..... **ON / CONFIRM if above 12,500 ft**

▶ Restart attempt practicable?

Yes – restart:

- Magnetos ..... **BOTH – check**
- Alternate induction air ..... **CHECK (automatic door; ALT AIR indication if induction icing)**
- Throttle ..... **1/4 OPEN**
- Starter ..... **ENGAGE if propeller has stopped**

▶ Engine restarted?

Yes:

- Power ..... **AS REQUIRED – apply slowly**
- Fuel pump ..... **LO if required**

Deferred: Engine restarted after in-flight failure – land as soon as possible, suspect fuel or induction system. ..... **NOTED**

No:

Continue with forced landing.

No – forced landing:

Forced landing without power:

- ATC / ELT ..... **MAYDAY, squawk 7700 (MFD nearest airport)**
- Seats / belts / harnesses ..... **SECURE; passengers briefed**
- Landing gear ..... **DOWN on firm surface (below 165 KIAS; emergency extension if hydraulic pump fails) / UP on rough terrain or water**
- Flaps ..... **DOWN when landing assured**
- Airspeed ..... **80–85 KIAS on final**
- Fuel selector ..... **OFF**
- Mixture ..... **IDLE CUTOFF**
- Magnetos ..... **OFF**
- Battery master / alternators ..... **OFF (after gear and flaps set)**
- Cabin door ..... **UNLATCH before touchdown**

Engine Fire in Flight

LAND IMMEDIATELY

- Fuel selector (memory) — **OFF**
- Mixture (memory) — **IDLE CUTOFF**
- Fuel pump (memory) — **OFF**
- Throttle (memory) — **CLOSED**
- Cabin heat / defroster / vents (memory) — **OFF / CLOSED**
- Oxygen — **OFF unless needed for smoke (oxygen feeds fire)**
- Airspeed — **INCREASE (up to 165 KIAS) to help extinguish fire**
- Landing gear — **DOWN while battery power remains (emergency extension if needed)**
- Forced landing — **EXECUTE (do not attempt restart)**

Engine Fire During Start

- Starter (memory) — **CONTINUE CRANKING**

▶ Engine starts?

Yes:

- Throttle — **1,700 RPM for a few minutes**
- Engine — **SHUT DOWN and inspect for damage**

No:

- Mixture — **IDLE CUTOFF**
- Throttle — **FULL OPEN**
- Fuel pump — **OFF**
- Fuel selector — **OFF**
- Starter — **CONTINUE CRANKING to draw fire into engine**
- Magnetos — **OFF**
- Battery master / alternators — **OFF**
- Fire extinguisher — **OBTAIN / USE**
- Aircraft — **EVACUATE**

Electrical / Cabin Fire and Smoke

LAND AS SOON AS POSSIBLE

- Battery master / alternator 1 & 2 (memory) — **OFF**
- Avionics master (memory) — **OFF**
- Cabin heat / defroster / vents — **OFF / CLOSED**
- Fire extinguisher — **ACTIVATE (then ventilate cabin)**
- Oxygen — **CREW MASKS ON if smoke persists (keep oxygen away from flames)**

With electrical power off the PFD and MFD are dark: fly on the standby instruments (backup battery).

▶ Fire out and electrical power required?

Yes:

- Circuit breakers — **CHECK – do not reset faulted**
- Battery master — **ON**
- Alternators / avionics / equipment — **ON one at a time, check for smoke**
- Deferred: Electrical fire – suspect circuits isolated, affected systems unavailable. — **NOTED**

No:

Landing gear hydraulic pump and flaps are electric: plan an emergency free-fall gear extension and a flaps-up landing.

- Deferred: Electrical power off – standby instruments only, emergency gear extension, flaps-up landing, no landing light or gear lights. — **NOTED**

Smoke evacuation:

- Fresh air vents / storm window — **OPEN**
- Descent — **TO lower altitude if practicable**

Emergency Descent

- Throttle (memory) — **CLOSED (reduce smoothly if time permits)**
- Propeller (memory) — **FULL FORWARD**
- Landing gear (memory) — **DOWN (below 165 KIAS)**
- Flaps (memory) — **10° (below 165 KIAS)**
- Airspeed (memory) — **165 KIAS (smooth air) / 133 KIAS (turbulence)**
- Oxygen — **ON if above 12,500 ft**
- Mixture — **ENRICH as required**
- ATC — **ADVISE – emergency descent**
- Level-off — **10,000 ft or MEA – restore power slowly; gear up below 130 KIAS if continuing**

Hypoxia Symptoms / Oxygen System Failure

Above 12,500 ft: headache, euphoria, tingling, blue lips, poor judgement, low oxygen flow or pressure

LAND AS SOON AS PRACTICAL

- Oxygen mask (memory) — **DON – oxygen ON, flow CHECK (mask in place of cannula)**
- Descent (memory) — **INITIATE immediately to 12,500 ft or below (10,000 ft preferred) – Emergency Descent if symptoms severe**
- Autopilot (memory) — **ON – altitude / heading, to protect against incapacitation**
- ATC — **ADVISE – request lower**
- Passengers — **CHECK responsive; masks on**
- Oxygen system — **CHECK pressure, flow indicators, connections, bottle valve ON, lines not kinked**

▶ Oxygen flow restored?

Yes:

- Altitude — **REMAIN below 12,500 ft until symptoms clear; monitor with pulse oximeter**
- Deferred: Hypoxia event – remain low, oxygen system suspect; medical check for occupants. — **NOTED**

No:

- Altitude — **REMAIN at or below 12,500 ft for the rest of the flight (10,000 ft with passengers)**
- Deferred: Oxygen system failed – no flight above 12,500 ft, replan route / altitude. — **NOTED**

PFD Failure – Revert to Standby Instruments

PFD blank, frozen, or red X's on attitude / airspeed / altitude

- Standby instruments (memory) — **FLY attitude, airspeed, altimeter (standby attitude indicator on backup battery)**
- Autopilot (memory) — **OFF if attitude or flight director lost (G1000 GFC 700 requires AHRS / PFD)**
- Display reversion — **SELECT (G1000: DISPLAY BACKUP – PFD data on MFD; Avidyne: MFD and standby instruments)**
- PFD circuit breaker / dimming — **CHECK (brightness, breaker in)**
- Navigation / communication — **USE MFD / second GPS-nav-com; ATC advised**
- IMC — **EXIT / AVOID; request VMC approach or vectors**

▶ PFD data recovered (on PFD or MFD reversion)?

Yes:

- Autopilot — **RE-ENGAGE if attitude source valid**
- Deferred: PFD fault – display reversion in use, maintenance required. — **NOTED**

No:

- Approach — **VISUAL preferred; standby instruments and GPS for a non-precision approach**
- Deferred: PFD failed – standby instruments only, no autopilot, visual approach preferred. — **NOTED**

Emergency Gear Extension (Free Fall)

Gear does not extend hydraulically / HYD PUMP light stays on

- Airspeed — **REDUCE to approx 100 KIAS (below 165 KIAS Vle; slower aids free fall)**
- Hydraulic pump circuit breaker — **PULL**
- Landing gear selector — **DOWN**
- Emergency gear extension lever — **PULL / EMERGENCY position (floor, between front seats) – hydraulic pressure released, gear free-falls**
- Aircraft — **YAW left and right to assist gear into locks**
- Landing gear indicator — **THREE GREEN**

Do not reset the hydraulic pump circuit breaker or move the gear selector up after an emergency extension. Leave the lever in the emergency position and the gear down.

▶ Three green obtained?

Yes:

- Deferred: Gear extended by free fall – do not retract; gear and hydraulic system inspection required. — **NOTED**

No:

Continue to Gear-Up / Partial Gear Landing.

Gear-Up / Partial Gear Landing

Gear cannot be confirmed down

- Runway — **LONGEST HARD SURFACE available, emergency services advised**
- Passenger briefing — **COMPLETE (belts tight, brace, exits)**
- Landing gear — **UP (or as it is – do not cycle)**
- Flaps — **DOWN (36°)**
- Approach — **NORMAL, 80–85 KIAS**
- Fuel pump — **OFF**
- Cabin door — **UNLATCHED**

On short final:

- Mixture — **IDLE CUTOFF**
- Fuel selector — **OFF**
- Magnetos — **OFF**
- Battery master / alternators — **OFF**
- Touchdown — **MINIMUM SPEED, slightly nose high, wings level**
- Aircraft — **EVACUATE once stopped**

Inadvertent Icing Beyond Capability

Ice accumulating faster than the ice protection clears it, or aircraft not approved for icing

LAND AS SOON AS PRACTICAL

- Pitot heat / stall warning heat (memory) ..... **ON**
- Propeller deice / windshield heat (memory) ..... **ON (if fitted)**
- Surface deice (boots) (memory) ..... **CYCLE at first sign of ice – repeat as required (if fitted)**
- Exit icing (memory) ..... **TURN BACK / CLIMB / DESCEND – ATC advised, request new altitude**

Power ..... **INCREASE to hold airspeed; manifold pressure monitored (alternate air door opens automatically on induction ice)**

Autopilot ..... **OFF – hand fly, note control feel and trim**

Airspeed ..... **MAINTAIN higher than normal; no abrupt pitch changes**

Cabin heat / defroster ..... **FULL ON for forward view**

▶ Residual ice on airframe for landing?

No:

- Ice protection ..... **OFF when clear**

Yes:

- Flaps ..... **10° maximum for landing**
- Approach speed ..... **INCREASE 10–15 KIAS (approx 95–100 KIAS); expect stall warning unreliable**
- Landing ..... **LONGER RUNWAY, minimal flare, no steep turns**
- Deferred: Residual airframe ice – flaps 10°, approach 95–100 KIAS, longer landing distance. ..... **NOTED**

Spin Recovery

Intentional spins prohibited – inadvertent spin entry

- Throttle (memory) ..... **IDLE**
- Ailerons (memory) ..... **NEUTRAL**
- Rudder (memory) ..... **FULL OPPOSITE to rotation**
- Control wheel (memory) ..... **FULL FORWARD (release back pressure)**
- Rudder (memory) ..... **NEUTRAL as rotation stops**

Recovery ..... **SMOOTH pull-out from dive; do not exceed 198 KIAS (Vne) or load limits**

Flaps / gear ..... **UP during recovery if extended**

Power ..... **RESTORE slowly once in level flight**

Prototype content. Verify against the operator's approved checklist and the POH/AFM before any use. AeroE Checklist 7f510494 · 2026-11-07